

AdventureWorks Sales & Customer Intelligence Platform

1. Project Overview

This project aims to analyze sales performance, customer behavior, and product performance using the AdventureWorks database. The objective is to generate actionable business insights and build interactive dashboards using SQL Server and Power BI.

2. Tools & Technologies Used

- SQL Server
 - SQL Server Management Studio (SSMS)
 - Power BI Desktop
 - Excel
 - AdventureWorks Database
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3. Business Problem

The management team wants to understand:

- Which products generate the highest revenue?
- Which territories contribute the most sales?
- Who are the top customers?
- How does revenue change over time?
- Which product categories drive business growth?

The goal is to support data-driven business decisions.

4. Dataset Description

Tables Used:

1. Sales.Customer
2. Sales.SalesOrderHeader
3. Sales.SalesOrderDetail
4. Production.Product

5. Production.ProductSubcategory
6. Production.ProductCategory
7. Sales.SalesTerritory

The dataset contains customer information, order details, product information, and sales territory data.

5. SQL Analysis Performed

Revenue by Product Category

Analyzed revenue contribution of each product category.

Revenue by Territory

Identified top-performing sales territories.

Top 10 Products

Analyzed products generating the highest revenue.

Top 10 Customers

Identified the most valuable customers.

Customer Segmentation

Segmented customers using revenue-based ranking techniques.

Monthly Revenue Trend

Analyzed sales growth patterns over time.

6. Power BI Dashboard Development

Executive Dashboard

KPIs:

- Total Revenue
- Total Orders
- Total Customers
- Total Products

Visualizations:

- Monthly Revenue Trend
 - Revenue by Category
 - Revenue by Territory
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Customer Analytics Dashboard

Visualizations:

- Top 10 Customers by Revenue
 - Customer Revenue Analysis
 - Customer Segmentation
 - Business Insights
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Product Analytics Dashboard

Visualizations:

- Revenue by Product Category
 - Top 10 Products by Revenue
 - Product Insights
 - Business Recommendations
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7. Key Business Insights

Product Analysis

- Bikes generated approximately 86% of total revenue.
- Mountain-200 product series emerged as the top-performing product family.
- Components contributed around 11% of total revenue.
- Accessories and Clothing generated relatively low revenue.

Territory Analysis

- Southwest was the highest revenue-generating territory.
- Canada and Northwest also showed strong performance.
- Germany generated the lowest revenue among top territories.

Customer Analysis

- Top customers generated close to 1 million revenue each.
- Revenue distribution indicates a healthy customer base.
- Business is not dependent on a single customer.

Sales Trend Analysis

- Revenue increased significantly between 2011 and 2013.
 - 2013 was the strongest performing year.
 - March 2014 recorded the highest monthly revenue.
 - June 2014 appeared to contain incomplete sales data.
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8. Business Recommendations

- Increase focus on Bike category products.
 - Expand inventory for high-performing Mountain-200 products.
 - Develop marketing campaigns for low-performing categories.
 - Improve sales strategies in underperforming territories.
 - Launch loyalty programs targeting high-value customers.
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9. Project Outcome

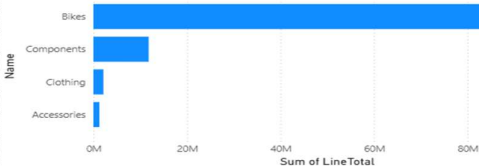
The project successfully transformed raw sales data into meaningful business insights through SQL analysis and interactive Power BI dashboards. The solution helps stakeholders monitor performance, identify growth opportunities, and make data-driven decisions.

10. Conclusion

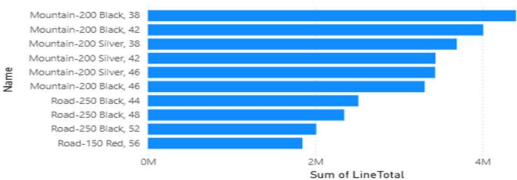
This project demonstrates end-to-end Business Intelligence skills including data extraction, SQL analysis, data modeling, dashboard development, and business insight generation using SQL Server and Power BI.

Product Performance Dashboard

Revenue by Category



Top 10 Products



Product Insights

- Bikes contribute 86% of total revenue.
- Mountain-200 series dominates sales.
- Components contribute 11% revenue.
- Accessories and Clothing contribute only 3%.

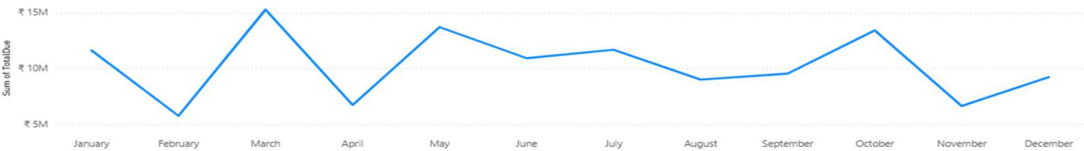
Business Recommendations

- Focus inventory on Bikes.
- Increase cross-selling of Accessories.
- Promote Clothing bundles.
- Expand Mountain-200 product line.

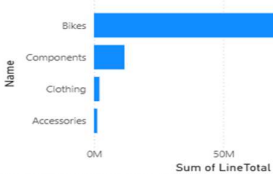
Executive Dashboard

Total Revenue ₹ 123.22M Total Orders 31.47K Total Customers 19.82K Total Products 504

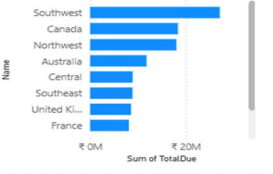
Sum of TotalDue by Month



Sum of LineTotal by Name



Sum of TotalDue by Name

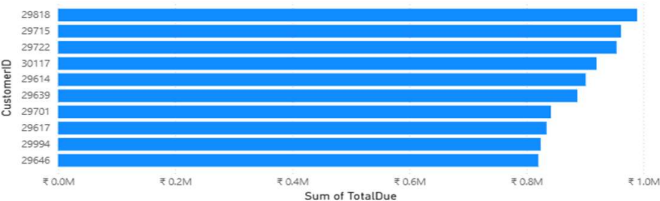


Name

- ☐ Australia
- ☐ Canada
- ☐ Central
- ☐ France
- ☐ Germany
- ☐ Northeast
- ☐ Northwest
- ☐ Southeast

Customer Analytics Dashboard

Top 10 Customers



Top Customers

CustomerID	Sum of TotalDue
29818	₹ 9,89,184.082
29715	₹ 9,61,675.8596
29722	₹ 9,54,021.9235
30117	₹ 9,19,801.8188
29614	₹ 9,01,346.856
29639	₹ 8,87,090.4106
29701	₹ 8,41,866.5522
29617	₹ 8,34,475.9271
29994	₹ 8,24,331.7682
29646	₹ 8,20,383.5466
29580	₹ 8,15,914.7534
Total	₹ 12,32,16,786.1159

Customer Segmentation Insights

- ♦ Platinum Customers Revenue: 3,119 - 989,184
- 🏆 Gold Customers Revenue: 607 - 3,119
- 🥈 Silver Customers Revenue: 61 - 607
- 🥉 Bronze Customers Revenue: 1.5 - 61

Key Business Insights

- Top customers generate nearly 1M revenue each.
- Revenue is well distributed among top customers.
- Business is not dependent on a single customer.
- Loyalty programs can increase customer retention.